Shawaz Imam

Software Engineer — Backend · Platform · Applied AI

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SUMMARY

Self-taught full-stack engineer (coding since 2018) with a backend and systems focus. I'm the sole engineer on a production multi-tenant SaaS (TypeScript, PostgreSQL, real-time AI voice) serving a property-management customer, and I build performance-critical services in Rust. I own the whole stack: schema, APIs, UI, CI/CD, infrastructure, and operations. I work test-first and rely on real-system verification, not mocks alone.

SKILLS

Languages	TypeScript, Rust, Python, SQL, JavaScript, Solidity, Bash, Nix
Backend	NestJS, Node.js, Fastify, Axum, Actix-web, tokio, Django/DRF, FastAPI; REST, WebSockets, webhooks
Frontend	React, Next.js, React Native/Expo, Tailwind, Redux; Playwright; i18n/RTL
Data	PostgreSQL (row-level security, migrations), Drizzle, SQLx, Prisma, Redis/BullMQ, SQLite, Qdrant
AI & voice	LLM tool use and fallback chains, MCP servers, RAG, Pipecat, streaming STT/TTS, Ollama, agentic dev workflows
Infra	GCP, Terraform, Docker, GitHub Actions, NixOS, systemd, Cloudflare, Firebase, Linux

EXPERIENCE

Software Engineer & Facilities Manager

Dec 2024 – Present

Property management company, Greater Boston

- Designed, built, and operate, as sole engineer, a **multi-tenant property-maintenance SaaS** with an AI front desk. It triages requests from phone, SMS, email, and web, opens work orders, dispatches vendors, and syncs to the customer's property-management system. TypeScript monorepo with 11 apps, 37 packages, 122 migrations, and about 1,900 commits.
- Enforced tenant isolation in the database: **forced Postgres row-level security** across all org-scoped tables, a least-privilege runtime role, and transaction-scoped context that prevents cross-org leaks through pooled connections. Verified with cross-org attack tests.
- Built a **real-time AI voice agent** (carrier media streams → streaming STT → LLM → TTS) with multi-vendor fallback, plus a synthetic-caller harness that places real calls in automated tests. The harness found stacked defects that had caused dropped calls.
- Implemented **safety-critical triage** (life-safety and seasonal no-heat floors that config can't lower) and an emergency dispatch cascade that pages on-call staff when every vendor has been tried.
- Integrated **Stripe subscriptions**: signed webhooks, protection against out-of-order events, and idempotency keys hashed from request parameters. A live sandbox contract suite (50 tests) caught three defects that mocked unit tests had passed.
- Built CI/CD: an 8-job GitHub Actions pipeline (secret scan, smoke boot, visual and accessibility checks, end-to-end tests) and a one-command deploy with rollback behind a Playwright gate. Found and fixed a critical auth-bypass CVE in a dependency, a SQL-injection advisory, and a token-rotation race.
- Built a **guarded LLM agent** (MCP server) for operational records. Writes follow propose → confirm → execute with single-use, payload-bound tokens, and the agent runs in a sandboxed service limited to one outbound host.

- Manage facilities for 25 properties, including 24/7 emergency response and vendor coordination. That daily contact with operations drives the product's requirements.

TypeScript · NestJS · Next.js · Expo · PostgreSQL · Drizzle · Redis/BullMQ · Python · Pipecat · Twilio/Telnyx · Stripe · Terraform · GCP · Cloudflare

Independent Software Engineer 2024 – Present

Rust systems, DeFi, and applied AI (personal projects)

- **Multi-chain liquidation engine** (Rust, about 21K lines, 320+ tests): an event-driven pipeline that simulates each trade in a local EVM (revm) before any RPC call. Includes a 15-state opportunity lifecycle, circuit breakers, Multicall3 batching, and a Solidity flash-loan contract tested with Foundry and deployed on 5 EVM chains.
- **B2B marketplace backend** (Rust/Axum/SQLx, 5 crates, 150+ tests including Testcontainers): separate Postgres schemas and a backend that is the only writer keep supplier identity hidden from buyers. Append-only money and audit records, Stripe deposits, TOTP MFA.
- **Infrastructure**: a 6-host NixOS fleet managed from one flake, with VM tests and CI, local GPU inference, a RAG index over my codebases, and fully local voice assistants.

ADDITIONAL EXPERIENCE

Self-Taught Automotive Mechanic 2016 – Present

- **Brakes:** calipers, pads, rotors, hoses, drum brakes, brake-line splicing, bleeding.
- **Suspension/steering:** ball joints, inner and outer tie rods, CV axles, wheel bearings, lug studs.
- **Engine/drivetrain:** transmission work, alternators, starters, spark plugs, exhaust, oil changes.
- **Electrical/interior:** subwoofer and amplifier wiring, head units, dash cams, headlights, headliner reupholstery, wheel refinishing.

EDUCATION

B.A., Interdisciplinary Studies (Applied Psychology), University of Massachusetts Amherst 2024

Full Stack Developer Certification, Clarusway 2023

Self-directed study: Python, JavaScript, HTML, Flask, Git, Linux (Codecademy and tutorials) 2018 – Present